XIAMEN UNIVERSITY MALAYSIA
厦門大學馬來西亞分校

COM105 Engineering Physics (I)

Lecture 8: Rotation of Rigid Bodies

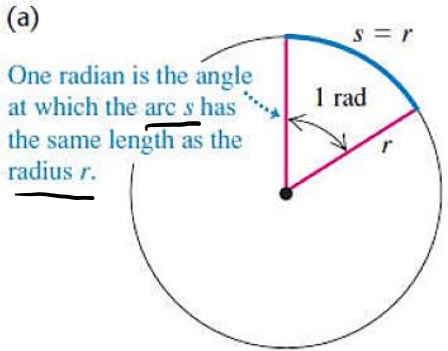
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In this chapter, you'll learn…

- how to describe the rotation of a rigid body in terms of angular coordinate, angular velocity, and angular acceleration.
- how to analyze rigid-body rotation when the angular acceleration is constant.
- the meaning of an object's moment of inertia about a rotation axis, and how it relates to rotational kinetic energy.
- how to calculate the moment of inertia of objects with various shapes, and different rotation axes.

Rotational Motion

- Rotation occurs at all scales, from the motions of electrons in atoms to the motions of entire galaxies. 银河
- To describe rotational motion, the most natural way to measure the angle is not in degrees, but in radians.



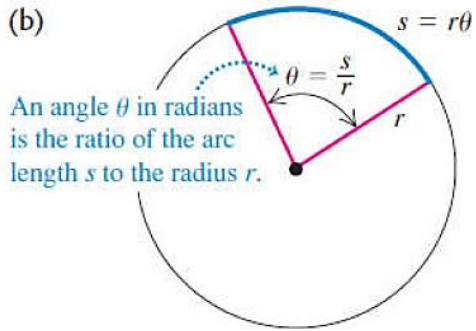
$\theta = 1$

$\Rightarrow$ 1 rad

$$\theta = \frac{s}{r} \quad \text{or} \quad s = r\theta \qquad (9.1)$$

$$1 \text{ rad} = \frac{360^\circ}{2\pi} = \frac{180^\circ}{\pi} \approx 57.30^\circ$$

$180^\circ = \pi$ rad, $90^\circ = \pi/2$ rad,



revolution per minute (rev/min or rpm)

Since 1 rev = 2π rad

1 rev/s = 2π rad/s

$$1 \text{ rev/min} = 1 \text{ rpm} = \frac{2\pi}{60} \text{rad/s}$$

Rotational Motion

- Average angular velocity

$$\bar{\omega} = \frac{\theta_2 - \theta_1}{t_2 - t_1} = \frac{\Delta\theta}{\Delta t} \quad (9.2)$$

- Instantaneous angular velocity

$$\omega = \lim_{\Delta t \to 0} \frac{\Delta\theta}{\Delta t} = \frac{d\theta}{dt} \quad (9.3)$$

- Average angular acceleration

$$\bar{\alpha} = \frac{\omega_2 - \omega_1}{t_2 - t_1} = \frac{\Delta\omega}{\Delta t} \quad (9.4)$$

- Instantaneous angular acceleration

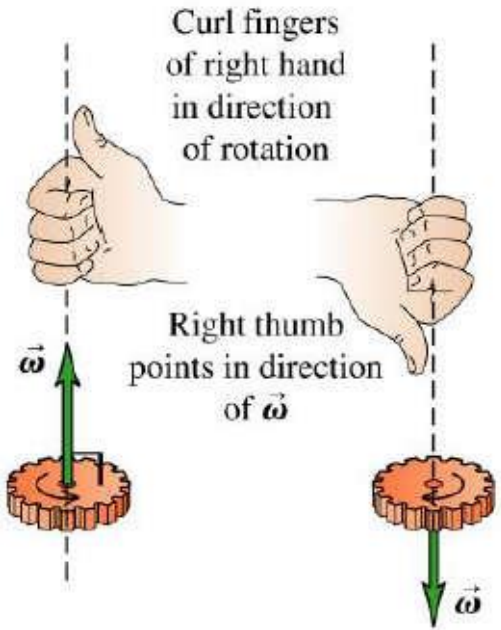
$$\alpha = \lim_{\Delta t \to 0} \frac{\Delta\omega}{\Delta t} = \frac{d\omega}{dt} \quad (9.5)$$

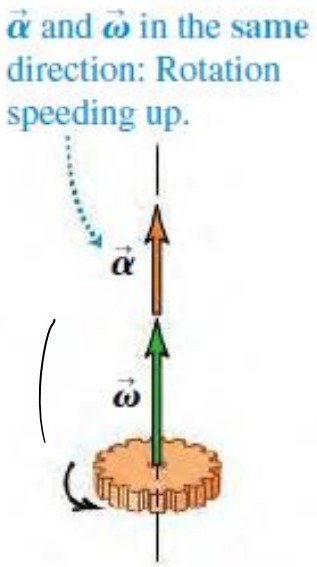
$$\alpha = \frac{d}{dt}\frac{d\theta}{dt} = \frac{d^2\theta}{dt^2} \quad (9.6)$$

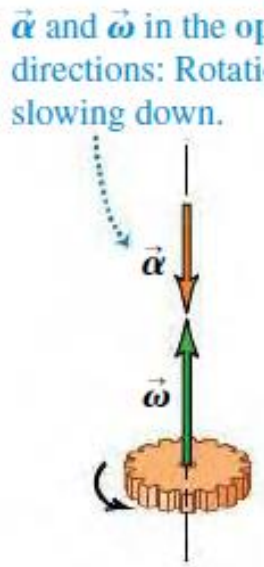
Angular Velocity and Acceleration As Vector

- **ω** and **α** are vectors.
- The direction of **ω** is determined by right-hand rule.
- The direction of **α** follows the definition $d\boldsymbol{\omega}/dt$.

The right-hand rule for the direction of the angular velocity vector **ω**. Reversing the direction of rotation reverses the direction of **ω**.







When the rotation axis is fixed, both the angular acceleration and angular velocity vectors lie along that axis.

Rotation With Constant Angular Acceleration

- Rotational Kinematics: Constant α

$$\alpha = \frac{d\omega}{dt} = \text{const.}$$

$$d\omega = \alpha dt$$

Integrating both sides and take $\omega = \omega_0$ at $t = 0$:

$$\omega = \omega_0 + \alpha t \quad (9.7)$$

$$\omega_{av} = \frac{\omega_0 + \omega}{2} \quad (9.8)$$

$$\omega_{av} = \frac{\theta - \theta_0}{t - 0} \quad (9.9)$$

Angular position at time t of a rigid body with constant angular acceleration

$$\theta - \theta_0 = \tfrac{1}{2}(\omega_{0z} + \omega_z)t \quad (9.10)$$

θ_0: Angular position of body at time 0

ω_{0z}: Angular velocity of body at time 0

ω_z: Angular velocity of body at time t

t: Time

$$\omega = \frac{d\theta}{dt}$$

$$d\theta = \omega dt = (\omega_0 + \alpha t)dt$$

Integrating both sides and take $\theta = \theta_0$ at $t = 0$

$$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2 \quad (9.11)$$

Eliminating t,

$$\omega^2 = \omega_0^2 + 2\alpha(\theta - \theta_0) \quad (9.12)$$

Rotation With Constant Angular Acceleration

- Kinematics: Angular vs. Linear
- For constant acceleration

线性中a.v.x的关系可以同理对应到角加速中

There is a one-to-one correspondence between:

$\theta \Longleftrightarrow x$

$\omega \Longleftrightarrow v$

$\alpha \Longleftrightarrow a$

这里的α加速度是单位t内角度的增加量

Straight-Line Motion with Constant Linear Acceleration		Fixed-Axis Rotation with Constant Angular Acceleration	
$a_x = \text{constant}$		$\alpha_z = \text{constant}$	
$v_x = v_{0x} + a_x t$	(2.8)	$\omega_z = \omega_{0z} + \alpha_z t$	(9.7)
$x = x_0 + v_{0x} t + \frac{1}{2} a_x t^2$	(2.12)	$\theta = \theta_0 + \omega_{0z} t + \frac{1}{2} \alpha_z t^2$	(9.11)
$v_x^2 = v_{0x}^2 + 2a_x(x - x_0)$	(2.13)	$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$	(9.12)
$x - x_0 = \frac{1}{2}(v_{0x} + v_x)t$	(2.14)	$\theta - \theta_0 = \frac{1}{2}(\omega_{0z} + \omega_z)t$	(9.10)

- If an object has a linear velocity v_x, the object as a whole is moving along the x-axis.

作为对比

- By contrast, if an object has an angular velocity ω_z, then it is rotating around the z-axis. We do not mean that the object is moving along the z-axis.

Rotation With Constant Angular Acceleration

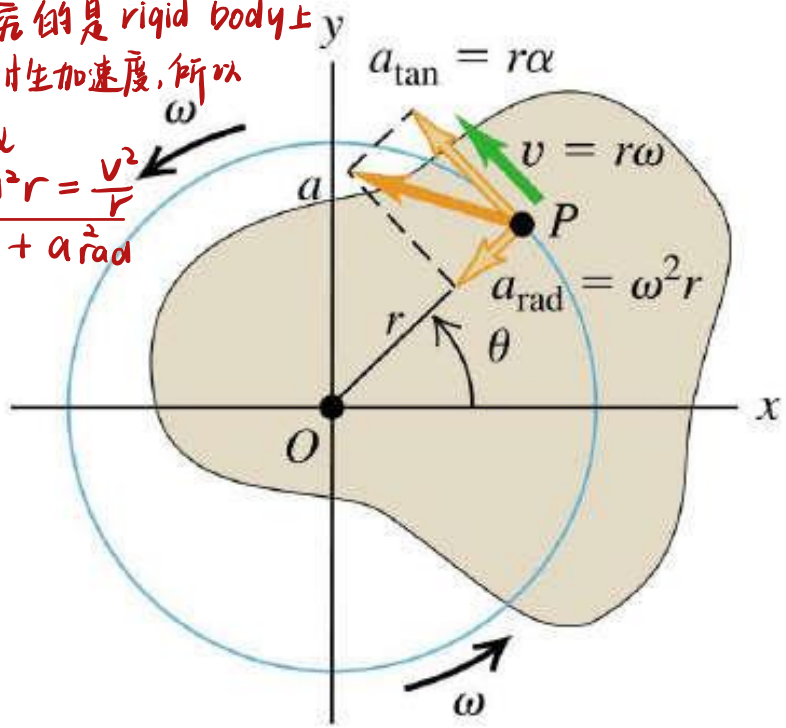
- When a rigid body rotates about a fixed axis, every particle in the body moves with the same angular speed ω:

此时，α 表示 ω 的加速度，但是我们在这里研究的是 rigid body 上一个点的线性加速度，所以

$a_{tan} = r\alpha$

$a_{rad} = \omega^2 r = \frac{v^2}{r}$

$a = \sqrt{a_{tan}^2 + a_{rad}^2}$



$$(s = r\theta)$$

$$v = \frac{ds}{dt} = r\frac{d\theta}{dt} = r\omega \qquad (9.13)$$

$$a_{tan} = \frac{dv}{dt} = r\frac{d\omega}{dt} = r\alpha \qquad (9.14)$$

Also:

$$a_{rad} = \frac{v^2}{r} = r\omega^2 \qquad (9.15)$$

Rotation With Constant Angular Acceleration

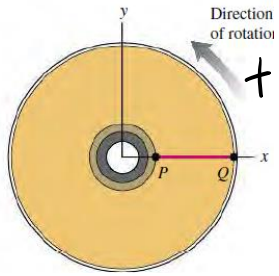
EXAMPLE 9.3 ROTATION WITH CONSTANT ANGULAR ACCELERATION

You have finished watching a movie on Blu-ray and the disc is slowing to a stop. The disc's angular velocity at $t = 0$ is 27.5 rad/s, and its angular acceleration is a constant $-10.0\ \text{rad/s}^2$. A line PQ on the disc's surface lies along the $+x$-axis at $t = 0$ (**Fig. 9.8**). (a) What is the disc's angular velocity at $t = 0.300$ s? (b) What angle does the line PQ make with the $+x$-axis at this time?

SOLUTION

IDENTIFY and SET UP: The angular acceleration of the disc is constant, so we can use any of the equations derived in this section (Table 9.1). Our target variables are the angular velocity ω_z and the angular displacement θ at $t = 0.300$ s. Given $\omega_{0z} = 27.5$ rad/s, $\theta_0 = 0$, and $\alpha_z = -10.0\ \text{rad/s}^2$, it's easiest to use Eqs. (9.7) and (9.11) to find the target variables.

9.8 A line PQ on a rotating Blu-ray disc at $t = 0$.



$t=0$
$\omega = 27.5$ rad/s
$\alpha = -10$ rad/s^2

EXECUTE: (a) From Eq. (9.7), at $t = 0.300$ s we have

$$\omega_z = \omega_{0z} + \alpha_z t = 27.5\ \text{rad/s} + (-10.0\ \text{rad/s}^2)(0.300\ \text{s})$$
$$= 24.5\ \text{rad/s}$$

(b) From Eq. (9.11),

$$\theta = \theta_0 + \omega_{0z} t + \tfrac{1}{2}\alpha_z t^2$$
$$= 0 + (27.5\ \text{rad/s})(0.300\ \text{s}) + \tfrac{1}{2}(-10.0\ \text{rad/s}^2)(0.300\ \text{s})^2$$
$$= 7.80\ \text{rad} = 7.80\ \text{rad}\left(\frac{1\ \text{rev}}{2\pi\ \text{rad}}\right) = 1.24\ \text{rev}$$

The disc has made one complete revolution plus an additiona 0.24 revolution—that is, 360° plus $(0.24\ \text{rev})(360°/\text{rev}) = 87°$ Hence the line PQ makes an angle of 87° with the $+x$-axis.

$\omega_3 = \omega_0 + \alpha \cdot t$
$= 27.5 - 10 \times 3 = 24.5$ rad/s
$\theta = \frac{1}{2}\alpha t^2 + \omega t = 7.8$ rad

$= 7.8 \text{ rad}\left(\frac{1 \text{ rev}}{2\pi \text{ rad}}\right) = 1.24 \text{ rev}$

Rotation With Constant Angular Acceleration

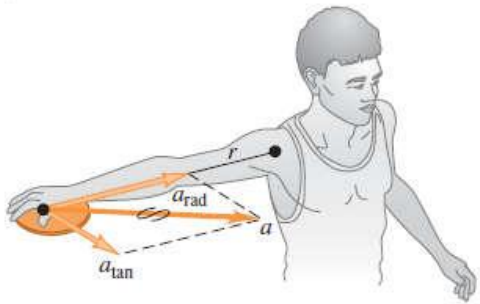
Example 9.4 **Throwing a discus**

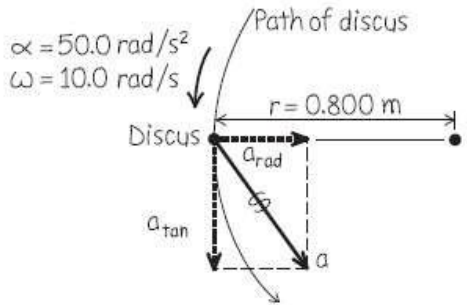
旋转 铁饼

An athlete whirls a discus in a circle of radius 80.0 cm (0.8 m). At a certain instant, the athlete is rotating at 10.0 rad/s and the angular speed is increasing at 50.0 rad/s^2. At this instant, find the tangential and centripetal components of the acceleration of the discus and the magnitude of the acceleration.

$\omega = 10 \text{ rad/s}$

$\alpha = 50 \text{ rad/s}^2$





$a_{tan} = r \cdot \alpha$

$a_{rad} = \omega^2 r$

EXECUTE: From Eqs. (9.14) and (9.15),

① $a_{tan} = r\alpha = (0.800 \text{ m})(50.0 \text{ rad/s}^2) = 40.0 \text{ m/s}^2$

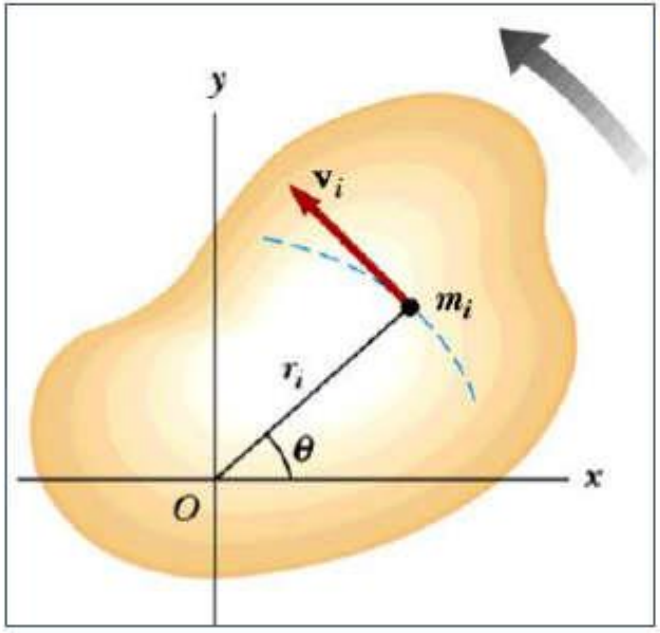
② $a_{rad} = \omega^2 r = (10.0 \text{ rad/s})^2(0.800 \text{ m}) = 80.0 \text{ m/s}^2$

Then

③ $a = \sqrt{a_{tan}^2 + a_{rad}^2} = 89.4 \text{ m/s}^2$

Energy in Rotational Motion

- Consider a rigid body as a collection of small particles rotating about a fixed z-axis at angular speed ω:



Kinetic energy of ith particle

$$K_i = \frac{1}{2}m_i v_i^2 = \frac{1}{2}m_i r_i^2 \omega^2$$

Total kinetic energy

$$K = \tfrac{1}{2}(m_1 r_1^2 + m_2 r_2^2 + \cdots)\omega^2 = \tfrac{1}{2}\left(\sum_i m_i r_i^2\right)\omega^2$$

$\frac{1}{2} m v^2 = \frac{1}{2} m \omega^2 r^2$

v = not same; ω = same; r = same ✓

引入惯性矩 the moment of inertia

Energy in Rotational Motion

- The moment of inertia, I, is obtained by multiplying the mass of each particle by the square of its distance from the axis of rotation and adding these products:

Moment of inertia of a body for a given rotation axis

$$I = m_1 r_1^2 + m_2 r_2^2 + \cdots = \sum_i m_i r_i^2 \qquad (9.16)$$

Masses of the particles that make up the body

Perpendicular distances of the particles from rotation axis

与 rotation axis 的垂直距离

- The rotational kinetic energy of a rigid body is therefore:

Rotational kinetic energy of a rigid body rotating around an axis

$$K = \frac{1}{2} I \omega^2 \qquad (9.17)$$

Moment of inertia of body for given rotation axis

Angular speed of body

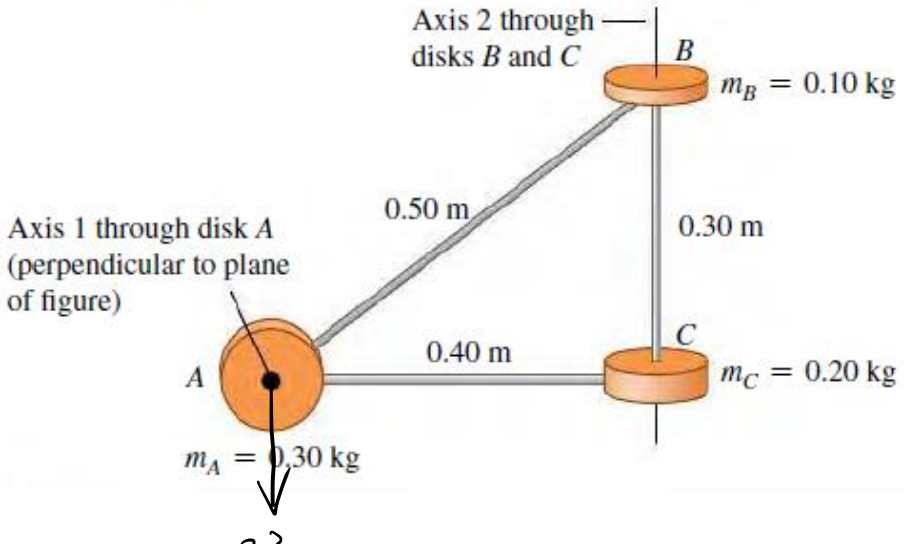
- The SI unit of I is kg.m^2.

Energy in Rotational Motion

EXAMPLE 9.6

A machine part (**Fig. 9.15**) consists of three disks linked by light-weight struts. (a) What is this body's moment of inertia about axis 1 through the center of disk A, perpendicular to the plane of the diagram? (b) What is its moment of inertia about axis 2 through the centers of disks B and C? (c) What is the body's kinetic energy if it rotates about axis 1 with angular speed $\omega = 4.0$ rad/s?

9.15 An oddly shaped machine part.



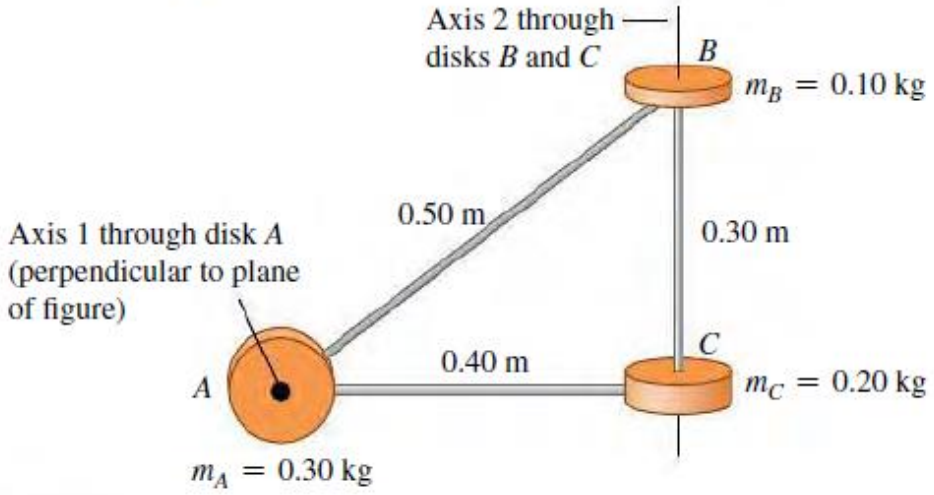
0.5

Energy in Rotational Motion

SOLUTION

IDENTIFY and SET UP: We'll consider the disks as massive particles located at the centers of the disks, and consider the struts as massless. In parts (a) and (b), we'll use Eq. (9.16) to find the moments of inertia. Given the moment of inertia about axis 1, we'll use Eq. (9.17) in part (c) to find the rotational kinetic energy.

9.15 An oddly shaped machine part.



EXECUTE: (a) The particle at point A lies *on* axis 1 through A, so its distance r from the axis is zero and it contributes nothing to the moment of inertia. Hence only B and C contribute in Eq. (9.16):

$$I_1 = \sum m_i r_i^2 = (0.10\ \text{kg})(0.50\ \text{m})^2 + (0.20\ \text{kg})(0.40\ \text{m})^2$$
$$= 0.057\ \text{kg} \cdot \text{m}^2$$

(b) The particles at B and C both lie on axis 2, so neither particle contributes to the moment of inertia. Hence only A contributes:

$$I_2 = \sum m_i r_i^2 = (0.30\ \text{kg})(0.40\ \text{m})^2 = 0.048\ \text{kg} \cdot \text{m}^2$$

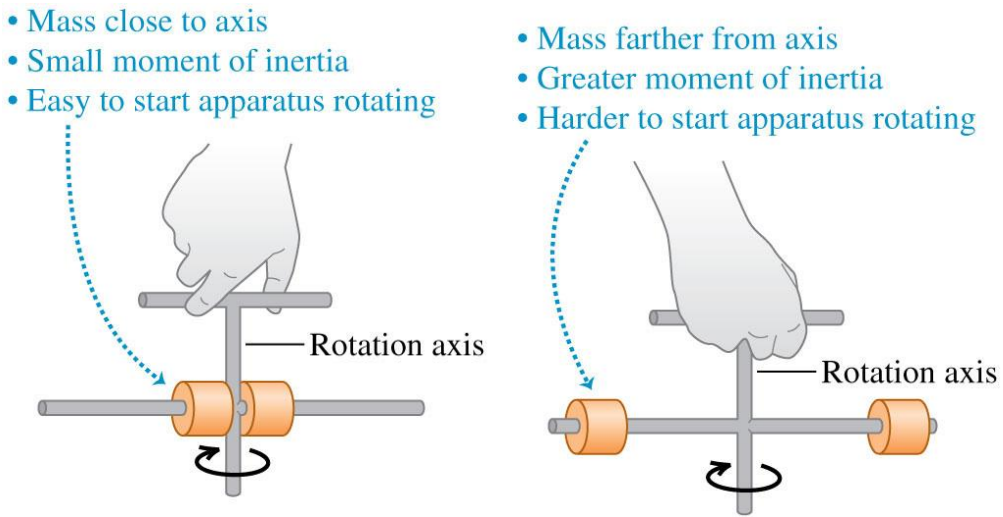
(c) From Eq. (9.17),

$$K_1 = \tfrac{1}{2} I_1 \omega^2 = \tfrac{1}{2}(0.057\ \text{kg} \cdot \text{m}^2)(4.0\ \text{rad/s})^2 = 0.46\ \text{J}$$

EVALUATE: The moment of inertia about axis 2 is smaller than that about axis 1. Hence, of the two axes, it's easier to make the machine part rotate about axis 2.

Moment of Inertia

- Here is an apparatus free to rotate around a vertical axis.
- To reduce the moment of inertia, lock the two equal-mass cylinders close to the center of the horizontal shaft.
- To increase the moment of inertia, lock the two equal-mass cylinders far from the center of the horizontal shaft.



Moment of Inertia of a Bird's Wing

- When a bird flaps its wings, it rotates the wings up and down around the shoulder.
- A hummingbird has small wings with a small moment of inertia, so the bird can move its wings rapidly (up to 70 beats per second).
- By contrast, the Andean condor has immense wings with a large moment of inertia, and flaps its wings at about one beat per second.





introduction

Moment of Inertia of Common Objects

- Mass is an intrinsic property of an object, but *I* depends on the physical arrangement of that mass, as well as the axis of rotation.
- If a rigid body is a continuous distribution (连续分布) of mass, like a solid cylinder or a solid sphere—it cannot be represented by a few point masses.
- Eq 9.16 then becomes an integral, imagine dividing the body into elements of mass *dm* that are very small, so that all points in a particular element are at essentially the same perpendicular distance from the axis of rotation, *r*.

$$I = \int r^2 \, dm \tag{9.20}$$

- For a 3D object, density (密度) $\rho = dm/dV$, where V = volume:

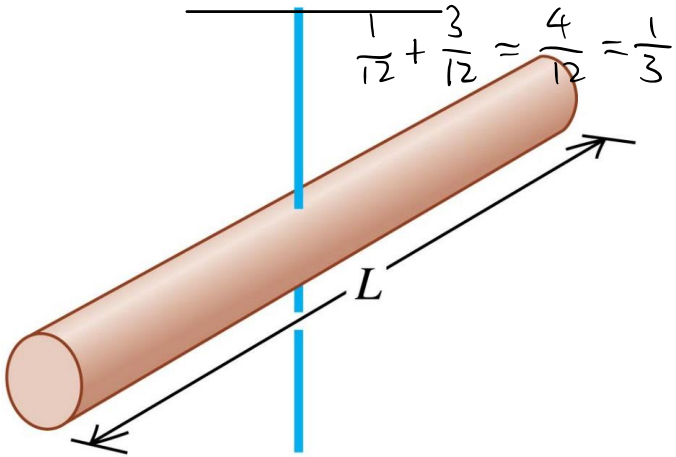
$$I = \int r^2 \rho \, dV = \rho \int r^2 \, dV \tag{9.21}$$

- In other words, the moment of inertia of any distribution of mass can be found by integrating over its volume

Moment of Inertia of Common Objects

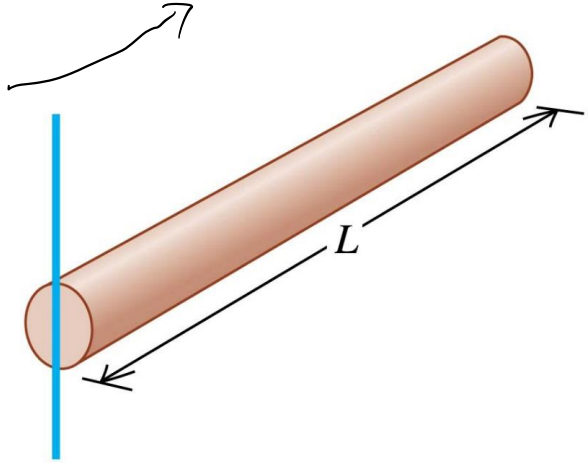
(a) Slender rod, axis through center

$$I = \frac{1}{12}ML^2$$



(b) Slender rod, axis through one end

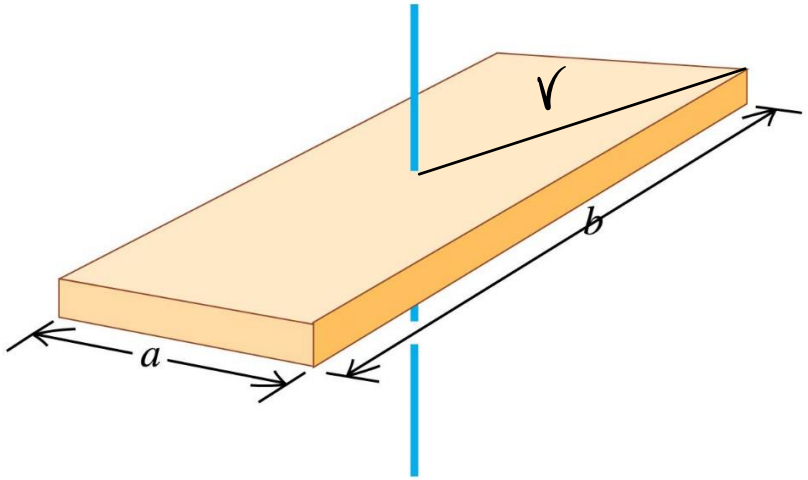
$$I = \frac{1}{3}ML^2$$



Moment of Inertia of Common Objects

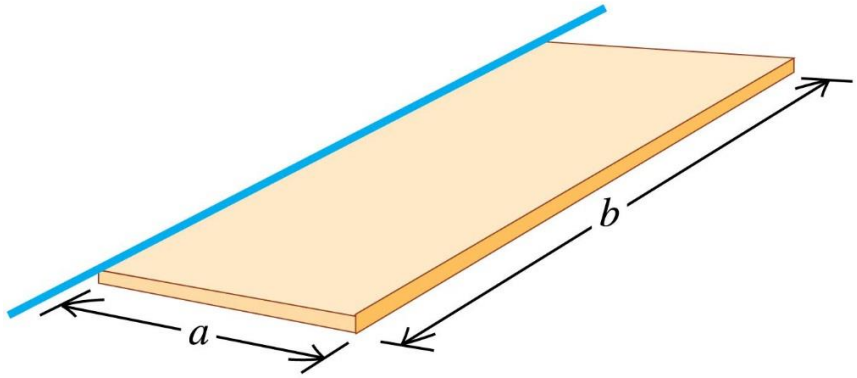
(c) Rectangular plate, axis through center

$$I = \frac{1}{12}M(a^2 + b^2)$$



(d) Thin rectangular plate, axis along edge

$$I = \frac{1}{3}Ma^2$$

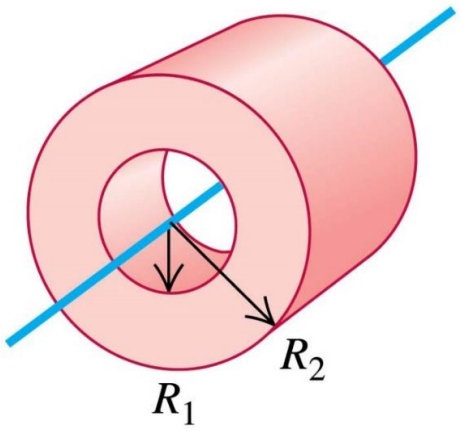


Moment of Inertia of Common Objects

中空 圓柱

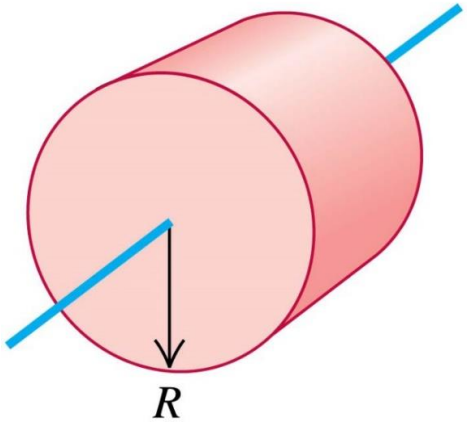
(e) Hollow cylinder

$$I = \frac{1}{2}M\left(R_1^2 + R_2^2\right)$$



(f) Solid cylinder

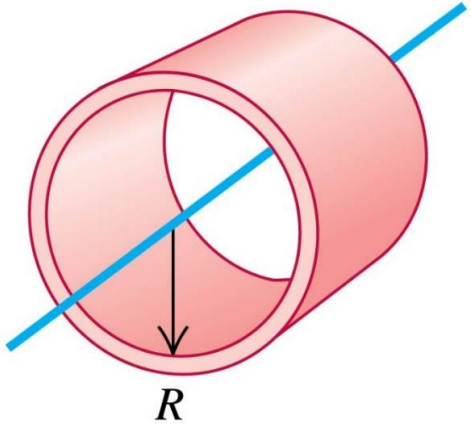
$$I = \frac{1}{2}MR^2$$



薄壁空心

(g) Thin-walled hollow cylinder

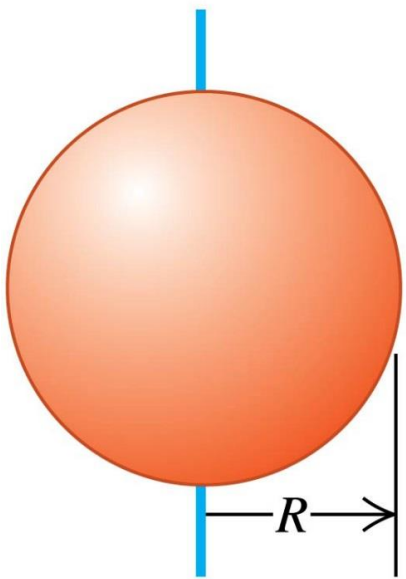
$$I = MR^2$$



Moment of Inertia of Common Objects

(h) Solid sphere sphere

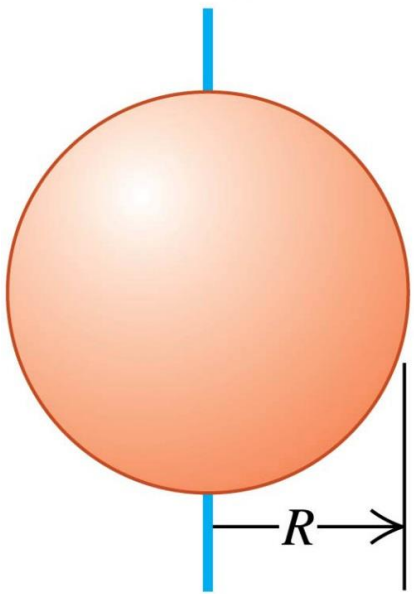
$$I = \frac{2}{5}MR^2$$



<

(i) Thin-walled hollow sphere

$$I = \frac{2}{3}MR^2$$



Moment of Inertia of Common Objects

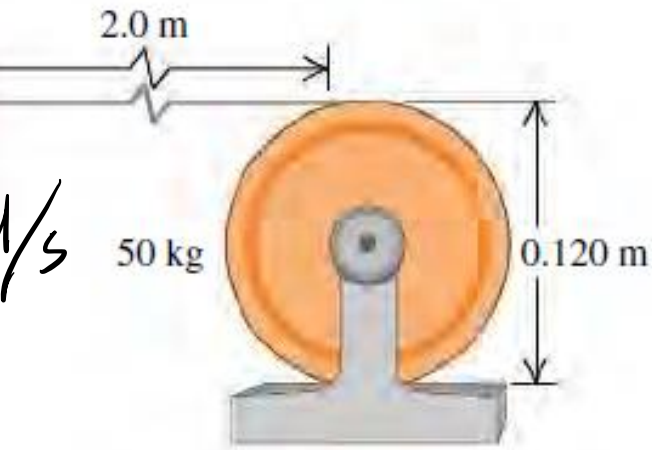
$9 \times 2 = W = 18\,J$

$K = \frac{1}{2} I w^2 = \frac{1}{2} \times \frac{1}{2} \times 50 \times 0.06^2 \times w^2$ (25)

$= 18$

$w = 20\ rad/s$

9.16 A cable unwinds from a cylinder (side view).



EXAMPLE 9.7 AN UNWINDING CABLE I

缠绕 不可拉伸

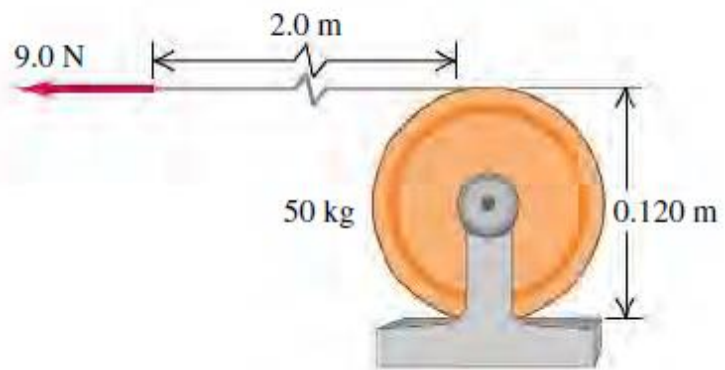
We wrap a light, nonstretching cable around a solid cylinder, of mass 50 kg and diameter 0.120 m, that rotates in frictionless bearings about a stationary horizontal axis (**Fig. 9.16**). We pull the free end of the cable with a constant 9.0-N force for a distance of 2.0 m; it turns the cylinder as it unwinds without slipping. The cylinder is initially at rest. Find its final angular speed and the final speed of the cable.

Moment of Inertia of Common Objects

SOLUTION $K_1 + U_1 + W_{others} = K_2 + U_2$

SET UP: Point 1 is when the cable begins to move. The cylinder starts at rest, so $K_1 = 0$. Point 2 is when the cable has moved a distance $s = 2.0$ m and the cylinder has kinetic energy $K_2 = \frac{1}{2}I\omega^2$. One of our target variables is ω; the other is the speed of the cable at point 2, which is equal to the tangential speed v of the cylinder at that point. We'll use Eq. (9.13) to find v from ω.

9.16 A cable unwinds from a cylinder (side view).



EXECUTE: The work done on the cylinder is $W_{other} = Fs = (9.0\text{ N})(2.0\text{ m}) = 18$ J. From Table 9.2 the moment of inertia is

$$I = \tfrac{1}{2}mR^2 = \tfrac{1}{2}(50\text{ kg})(0.060\text{ m})^2 = 0.090\text{ kg}\cdot\text{m}^2$$

(The radius R is half the diameter.) From Eq. (7.14), $K_1 + U_1 + W_{other} = K_2 + U_2$, so

$$0 + 0 + W_{other} = \tfrac{1}{2}I\omega^2 + 0$$

$$\omega = \sqrt{\frac{2W_{other}}{I}} = \sqrt{\frac{2(18\text{ J})}{0.090\text{ kg}\cdot\text{m}^2}} = 20\text{ rad/s}$$

From Eq. (9.13), the final tangential speed of the cylinder, and hence the final speed of the cable, is

$$v = R\omega = (0.060\text{ m})(20\text{ rad/s}) = 1.2\text{ m/s}$$

EVALUATE: If the cable mass is not negligible, some of the 18 J of work would go into the kinetic energy of the cable. Then the cylinder would have less kinetic energy and a lower angular speed than we calculated here.

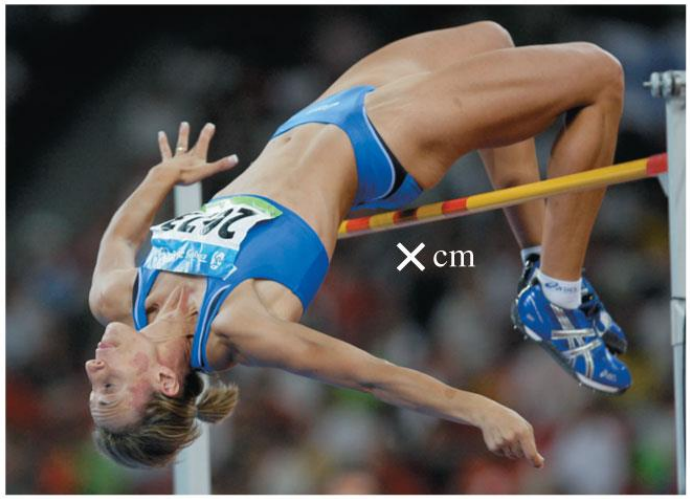
Gravitational Potential Energy of an Extended Object

伸展的物体要知道重心离地高度

- The gravitational potential energy of an extended object is the same as if all the mass were concentrated at its center of mass:

$$U = Mgy_{cm} \qquad (9.18)$$

- This athlete arches her body so that her center of mass passes under the bar.
- This technique requires a smaller increase in gravitational potential energy than straddling the bar.



Parallel-Axis Theorem

平行轴定理

- A body doesn't have just one moment of inertia. In fact, it has infinitely many, because there are infinitely many axes about which it might rotate.
- The moment of inertia about any axis parallel to and at distance d away from the axis that passes through the center of mass is:

Parallel-axis theorem:
Moment of inertia of a body for a rotation axis through point P

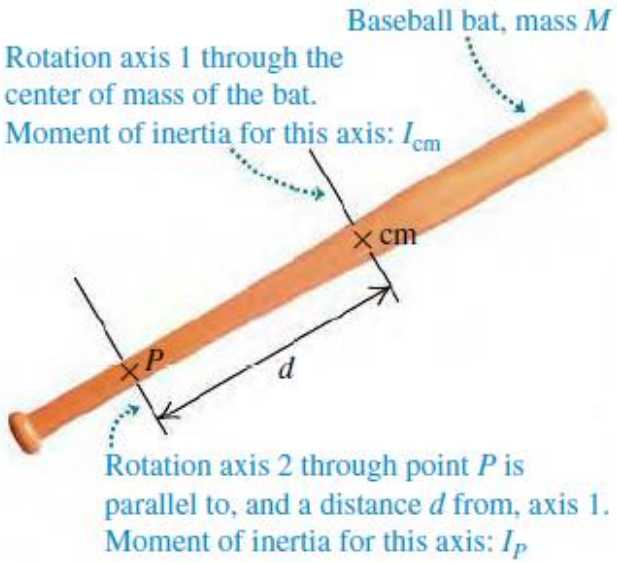
$$I_P = I_{cm} + Md^2 \quad (9.19)$$

I_{cm}: Moment of inertia of body for a parallel axis through center of mass

M: Mass of body

d: Distance between two parallel axes

重心处的 axis



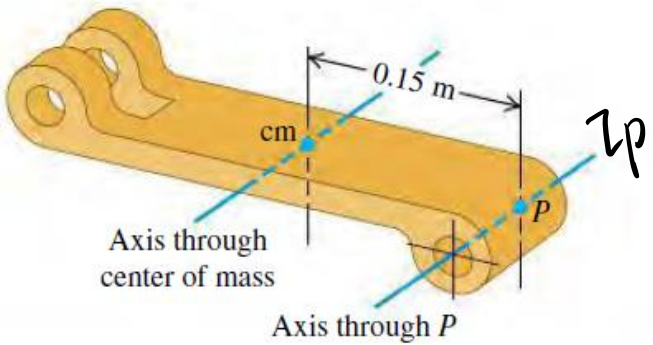
- This theorem works for any solids and shapes.

Parallel-Axis Theorem

EXAMPLE 9.9 USING THE PARALLEL-AXIS THEOREM

A part of a mechanical linkage (**Fig. 9.21**) has a mass of 3.6 kg. Its moment of inertia I_P about an axis 0.15 m from its center of mass is $I_P = 0.132\ \text{kg} \cdot \text{m}^2$. What is the moment of inertia I_{cm} about a parallel axis through the center of mass?

9.21 Calculating I_{cm} from a measurement of I_P.



3.6kg

I_P

$I_P = I_{cm} + Md^2$

$I_{cm} = I_P - Md^2$

$= 0.132 - 3.6 \times 0.15^2$

$= 0.051\ \text{kg} \cdot \text{m}^2.$

SOLUTION

IDENTIFY, SET UP, and EXECUTE: We'll determine the target variable I_{cm} by using the parallel-axis theorem, Eq. (9.19). Rearranging the equation, we obtain

$$I_{cm} = I_P - Md^2 = 0.132\ \text{kg} \cdot \text{m}^2 - (3.6\ \text{kg})(0.15\ \text{m})^2$$
$$= 0.051\ \text{kg} \cdot \text{m}^2$$

EVALUATE: As we expect, I_{cm} is less than I_P; the moment of inertia for an axis through the center of mass has a lower value than for any other parallel axis.

Summary

- Rotational motion of Rigid Bodies has been studied.
  - Basic concepts and principles are discussed with detailed analysis and examples
    1. **Rotational Motion**
    2. **Angular Velocity and Acceleration As Vector**
    3. **Rotation with Constant Angular Acceleration**
    4. **Energy in Rotational Motion**
    5. **Moment of Inertia**
    6. **Gravitational Potential Energy of an Extended Object**
    7. **Parallel-Axis Theorem**
- Textbook
  - Chapter 9：Rotational Motion of Rigid Bodies in the textbook.